

Shyamoli Textile Engineering College
Department of Computer Science and Engineering
CSE-4237: Digital Image Processing

Lec-3: Image Enhancement in the Spatial Domain

Digital Image Processing by Rafael C. Gonzalez and Richard Eugene Woods (Chapter-3)

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1. Spatial Domain

The term spatial domain refers to the aggregate of pixels composing an image. Spatial domain methods are procedures that operate directly on these pixels. Spatial domain processes will be denoted by the expression

$$g(x, y) = T[f(x, y)]$$

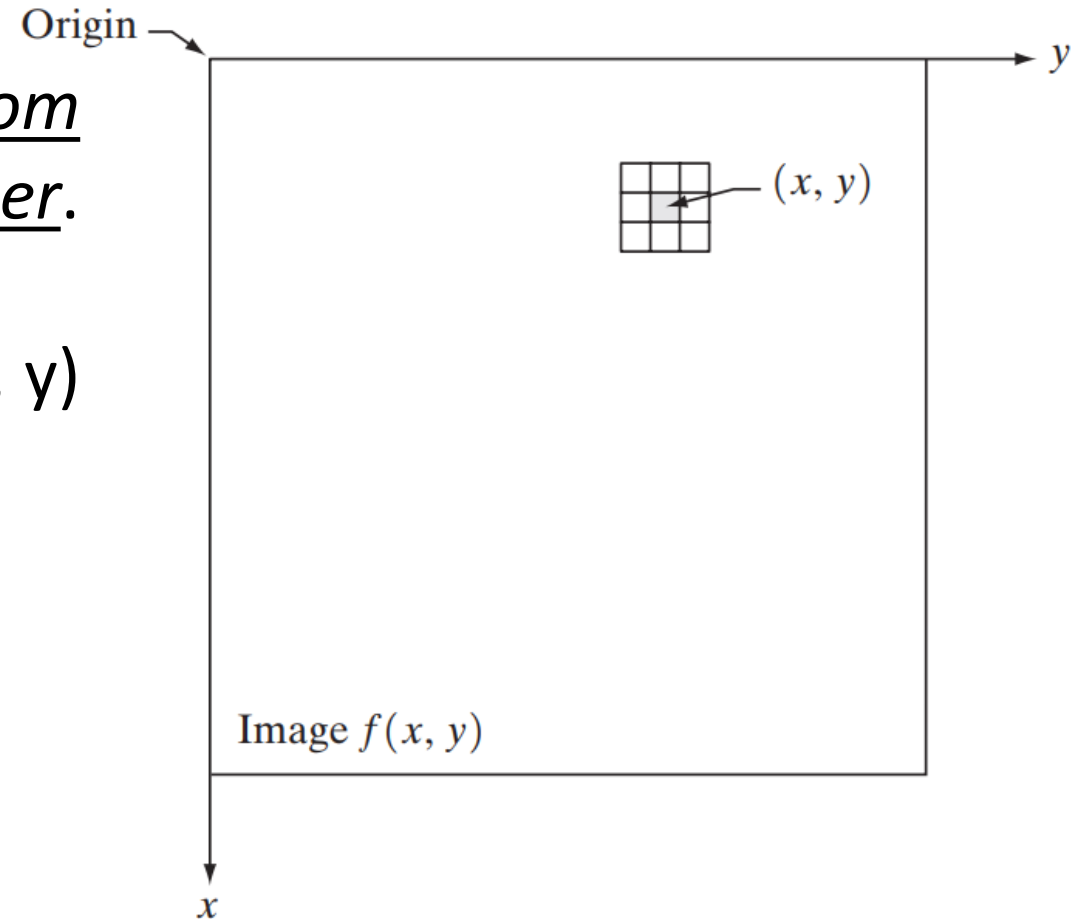
where $f(x, y)$ is the input image, $g(x, y)$ is the processed image, and T is an operator on f , defined over some neighborhood of (x, y) .

In addition, T can operate on a set of input images, such as performing the pixel-by-pixel sum of K images for noise reduction.

The principal approach in **defining a neighborhood about a point (x, y)** is to use a square or rectangular sub image area centered at (x, y) .

The center of the sub image is moved from pixel to pixel starting, say, at the top left corner.

The operator T is applied at each location (x, y) to yield the output, g , at that location.



A 3 X 3 neighborhood about
a point (x, y) in an image ₃

The simplest form of T is when the neighborhood is of size 1×1 (that is, a single pixel).

In this case, g depends only on the value of f at (x, y) , and T becomes a gray-level (also called an intensity or mapping) transformation function of the form

$$s = T(r)$$

r and s are variables denoting, respectively, the gray level of $f(x, y)$ and $g(x, y)$ at any point (x, y) .

1.1 Contrast Stretching

Contrast stretching is an Image Enhancement method which attempts to improve an image by stretching the range of intensity values.

❑ Intensity Levels (r):

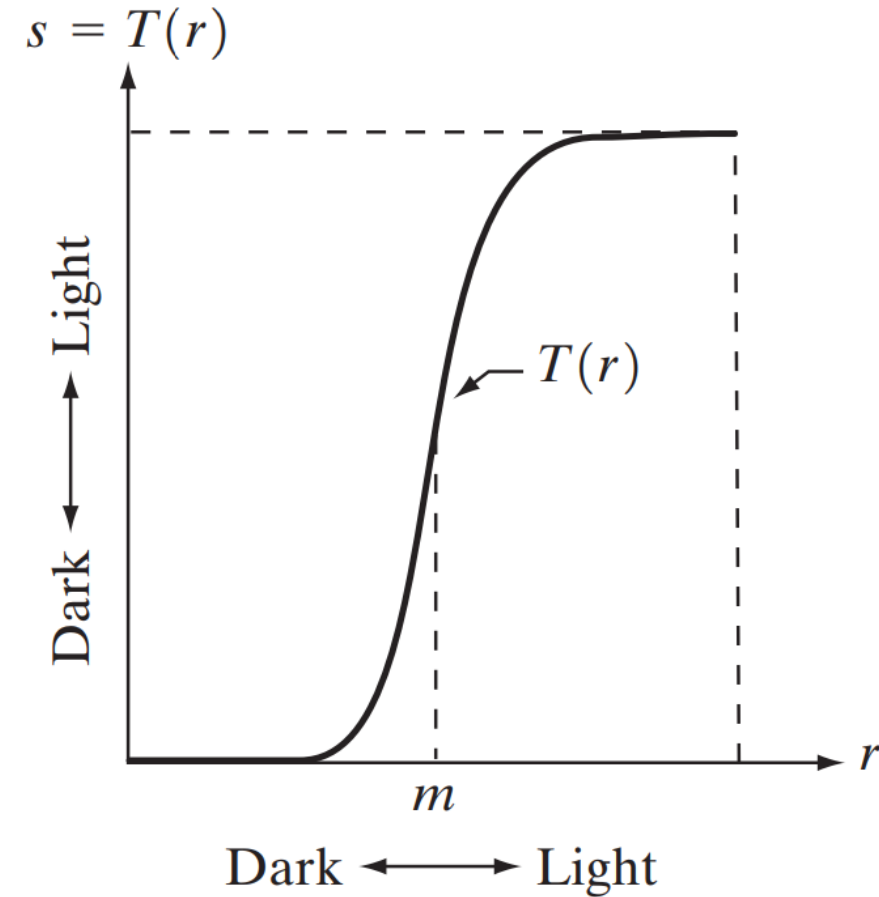
- ✓ Represent pixel intensity values in the original image.
- ✓ Range: Typically between 0 (black) and 255 (white) for an 8-bit image.

❑ Transformed Levels (s):

New intensity values after applying the transformation function $T(r)$.

❑ Threshold (m):

A midpoint intensity value around which the contrast adjustment is made.



Function $T(r)$:

This maps input intensity levels (r) to output levels (s).

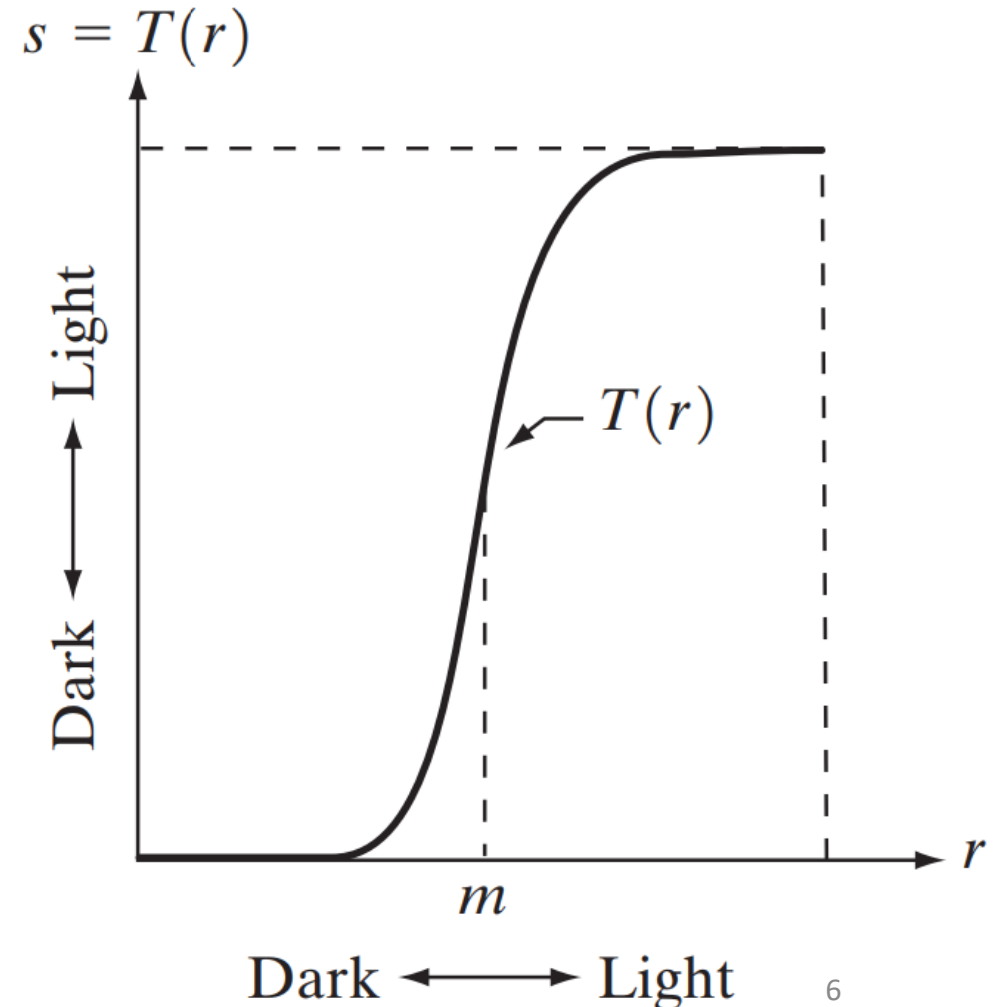
Effect of $T(r)$:

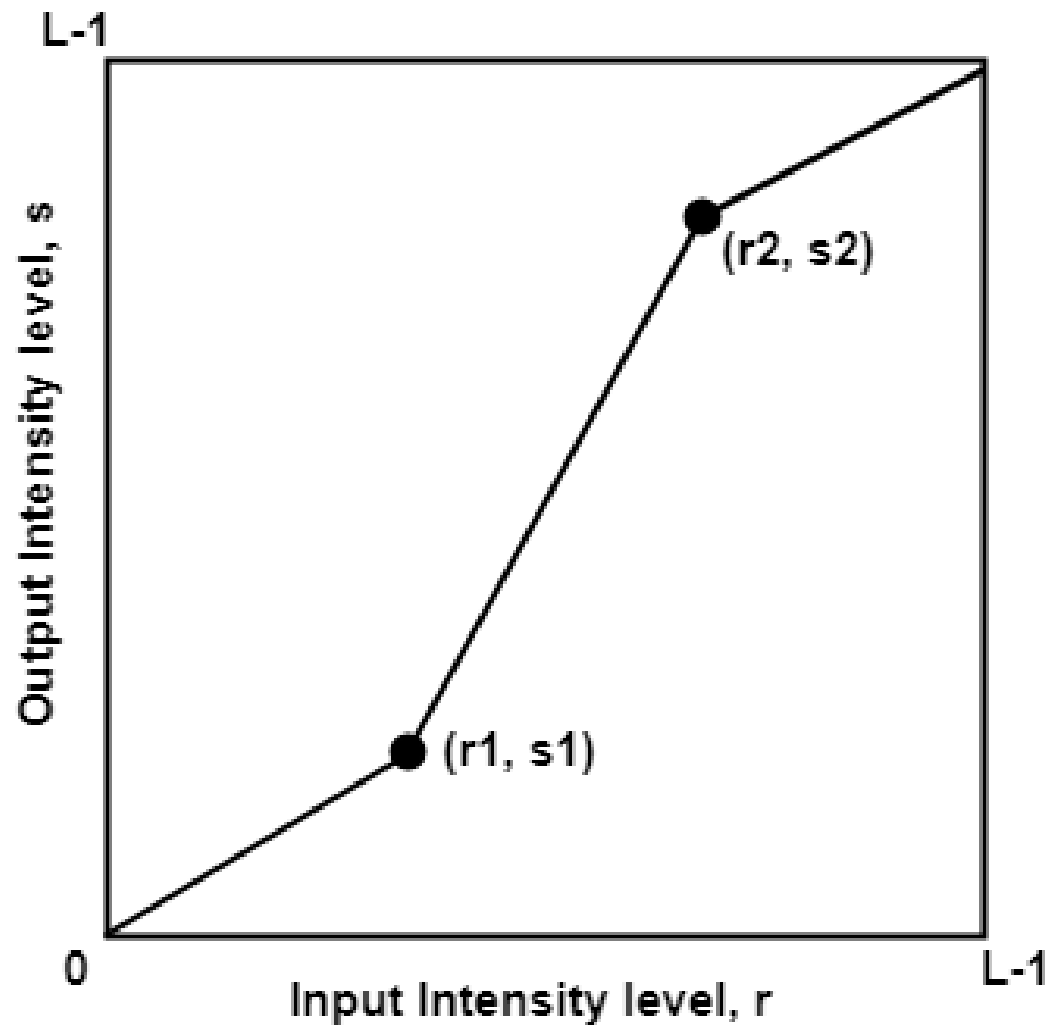
For $r < m$:

Input values below the midpoint (m) are compressed into a smaller range, making them appear darker.

For $r > m$:

Input values above the midpoint (m) are stretched to a wider range, making them appear brighter.





Original



Pixel range: 79-136

Contrast Stretched



Pixel range: 0-255

1.2 Thresholding Function ($T(r)$):

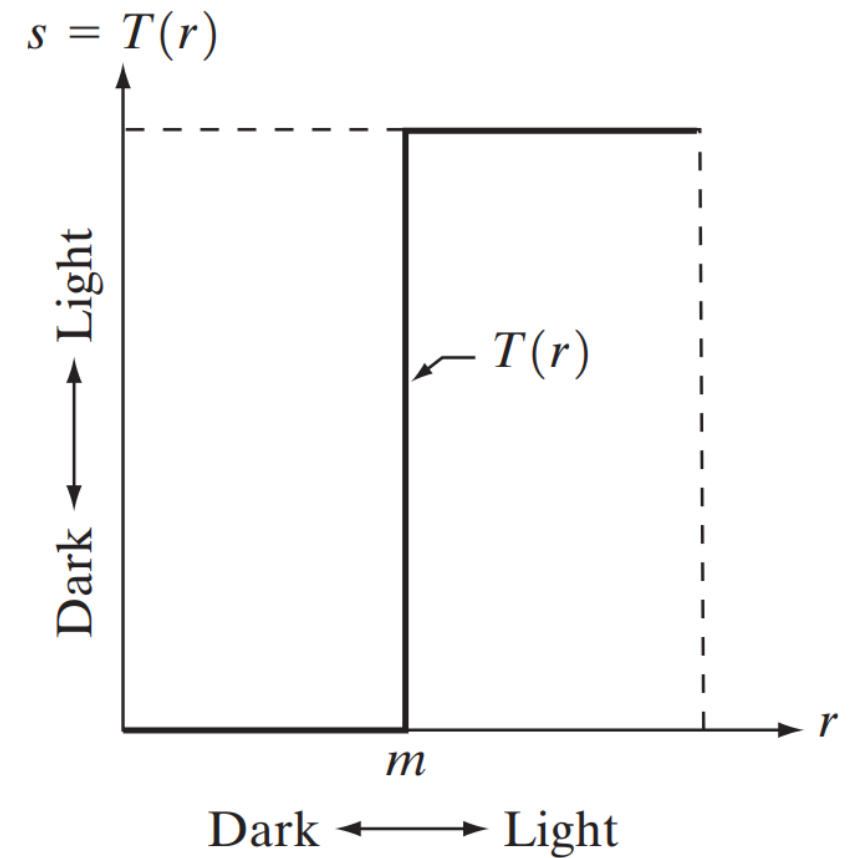
A thresholding function is a specific type of transformation that maps gray levels in an image (r) to either one of two levels (e.g., black or white) based on a threshold m .

In essence, it converts a grayscale image into a binary image.

Two-Level Binary Image:

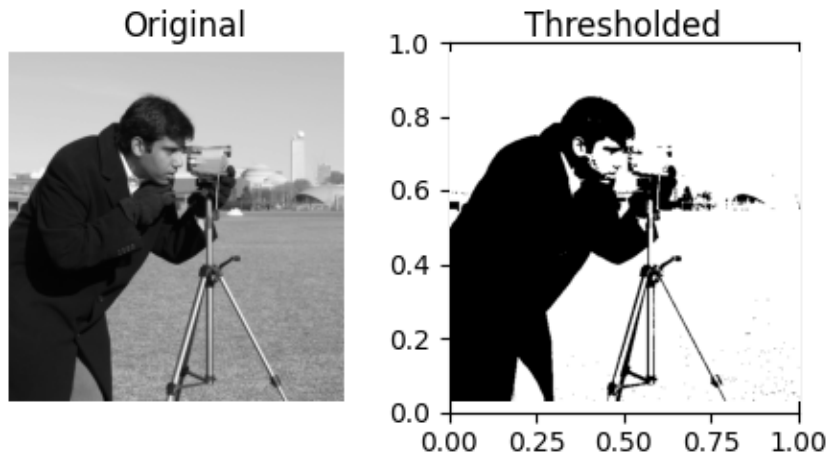
The output intensity values (s) of a binary image are limited to two levels:

typically 0 (black) and 255 (white).

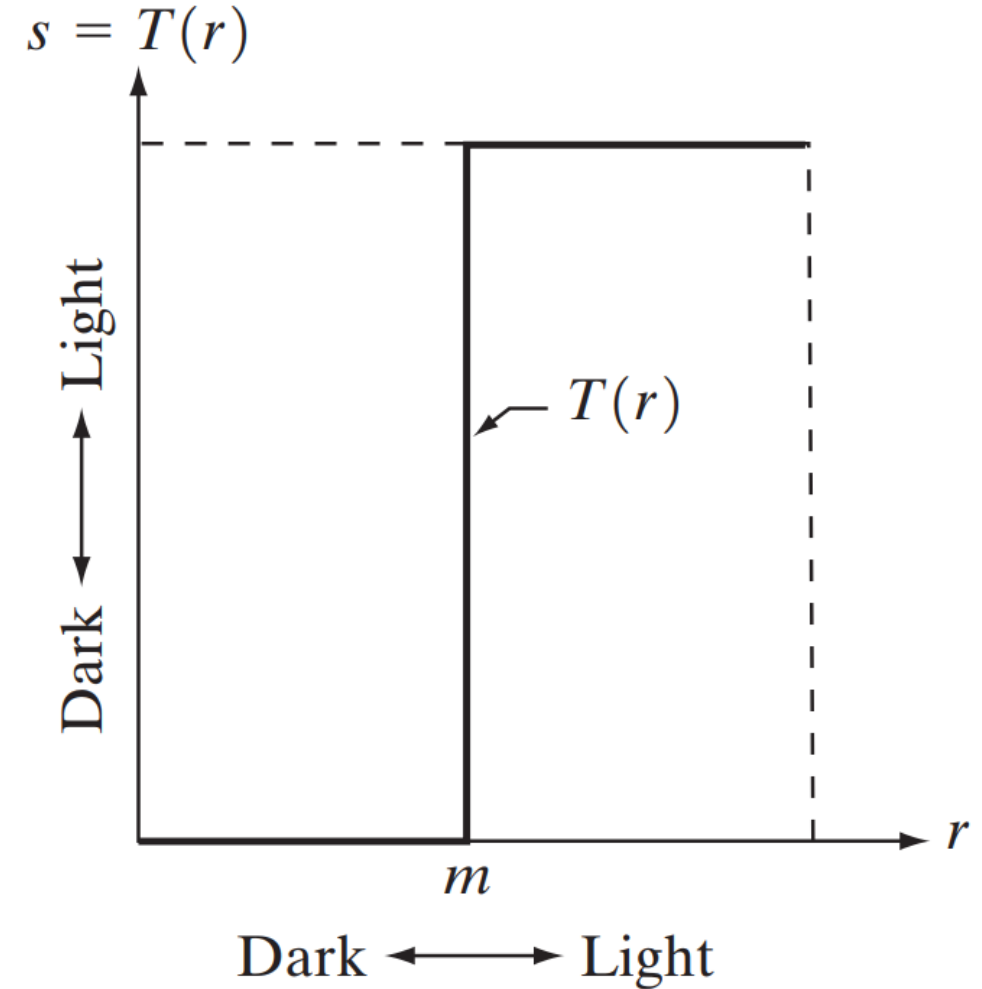


For a given threshold m :

$$T(r) = \begin{cases} 0, & \text{if } r < m \\ 255, & \text{if } r \geq m \end{cases}$$



- ❑ All pixel values below m are set to black.
- ❑ All pixel values equal to or above m are set to white.



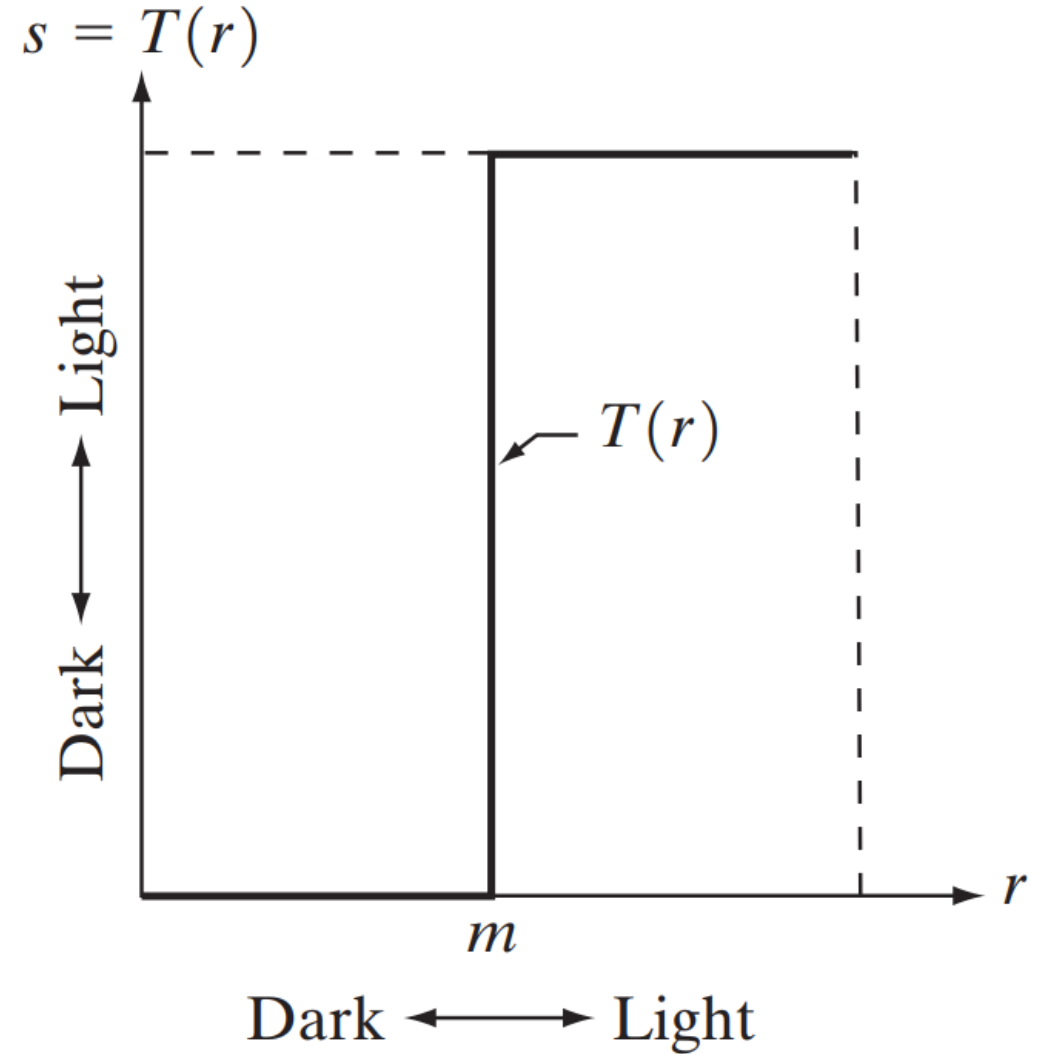
1.2.1 Thresholding Example:

Original Image:

A grayscale image with intensity levels ranging from 0 (black) to 255 (white).

Threshold $m=128$:

- ❑ If a pixel's intensity r is less than 128, it becomes black (0).
- ❑ If $r \geq 128$, it becomes white (255).



Input: Pixel intensities:

[50, 120, 130, 200, 250]

Output (Binary Image):

[0, 0, 255, 255, 255]

1.2.2 Applications:

☐ **Segmentation:**

To separate objects from the background, e.g., isolating text in scanned documents.

☐ **Edge Detection:**

Simplifies detecting object boundaries.

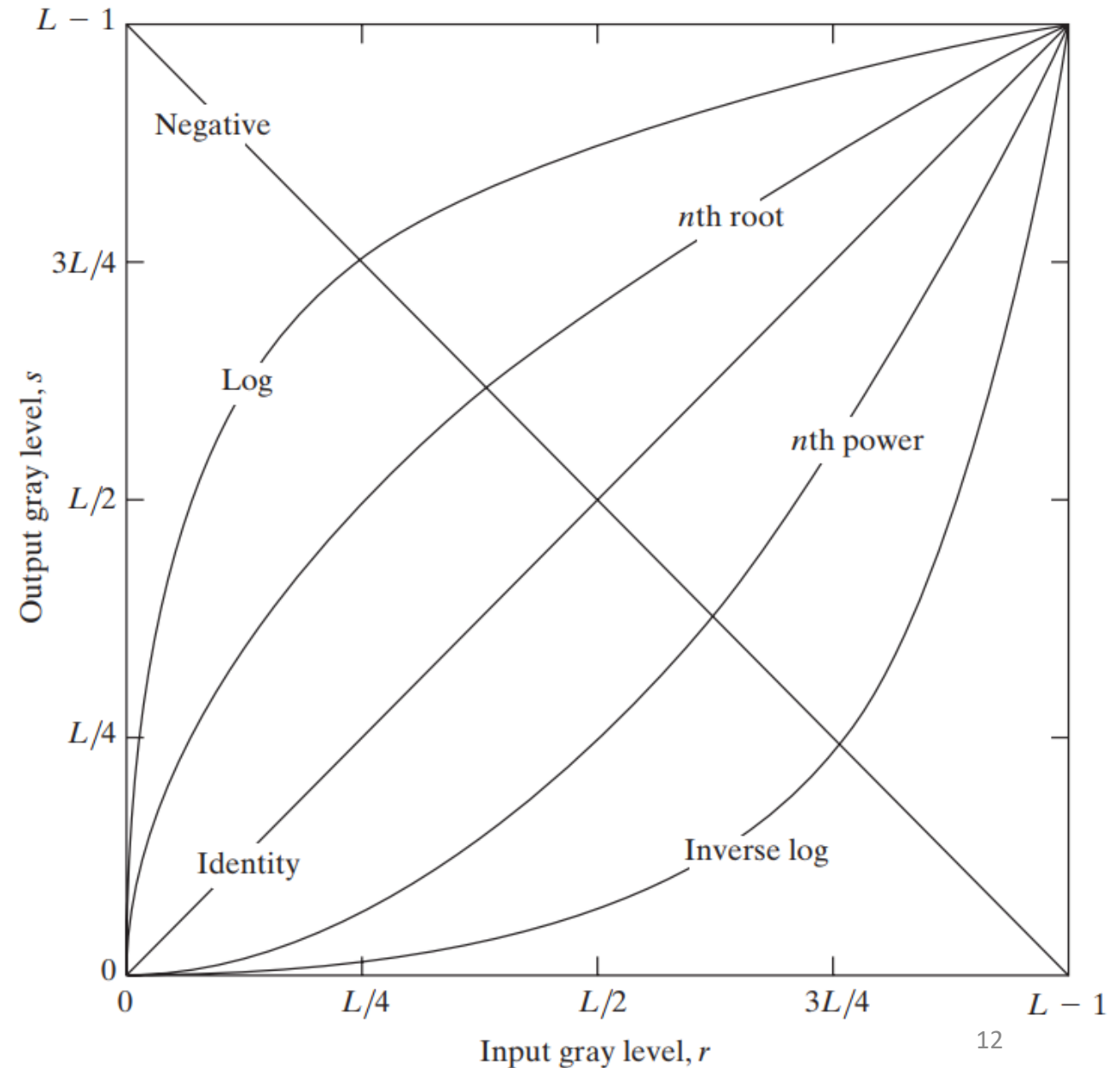
☐ **Simplified Analysis:**

Converts complex gray-level data into a manageable binary format for feature extraction.

2. Some Basic Gray Level Transformations

Three basic types of functions used frequently for image enhancement:

- ❑ linear (negative and identity transformations),
- ❑ logarithmic (log and inverse-log transformations), and
- ❑ power-law (nth power and nth root transformations).



2.1 Linear Transformations:

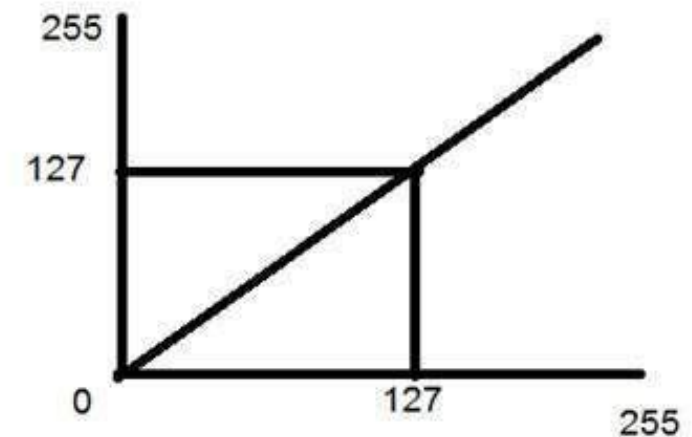
Linear transformations change the intensity of an image using a direct proportional relationship.

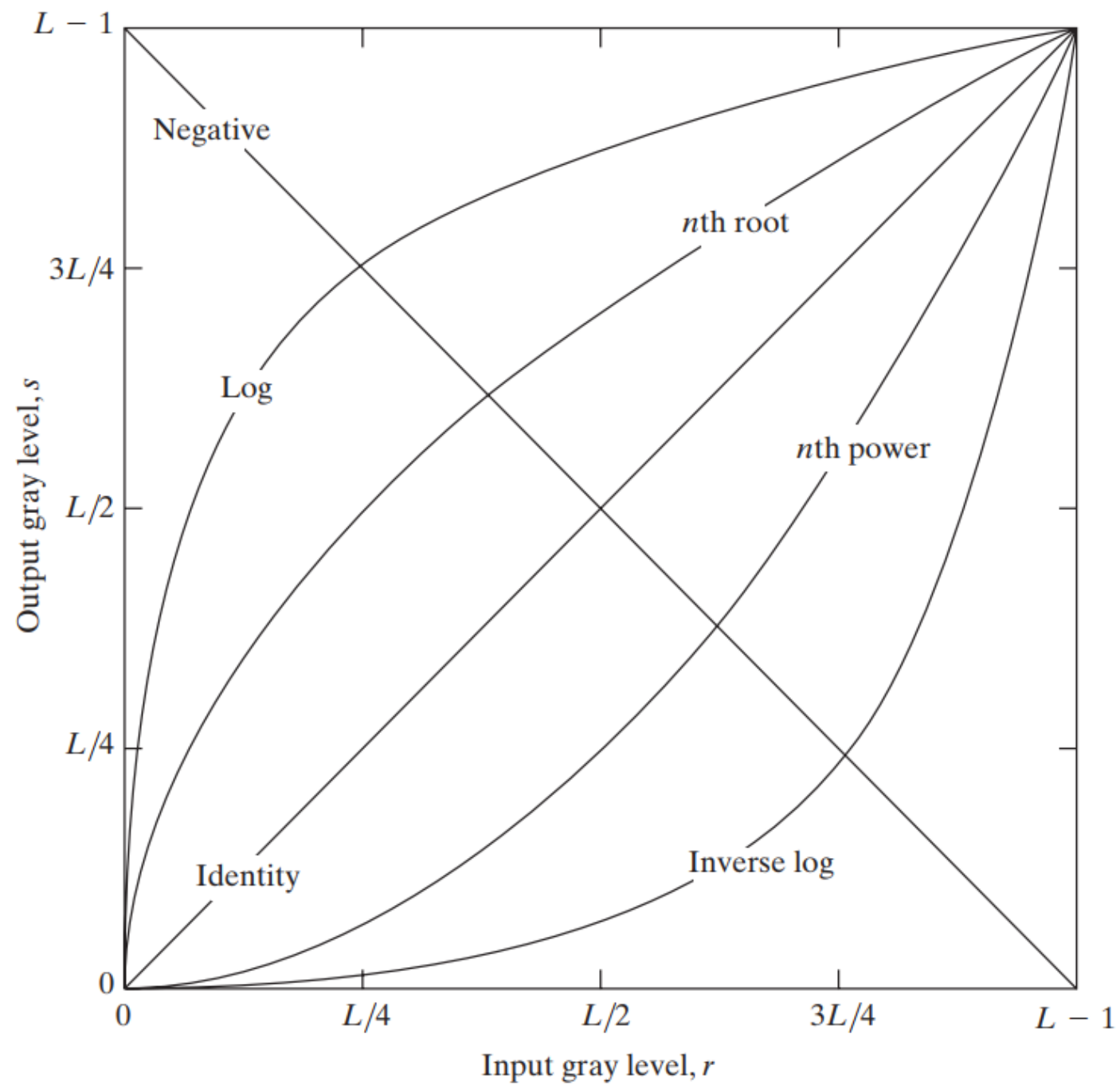
a) Identity Transformation:

The output pixel intensity is the same as the input pixel intensity. No change is made to the image.

Equation: $s = r$

Example: A pixel with intensity 127 remains 127.





b) Negative Transformation:

Converts an image to its negative by inverting the intensity values.

Equation: $s = L - 1 - r$

L: Maximum intensity level (e.g., 256 for an 8-bit image); r: Input pixel intensity.

Effect: Darker regions become bright, and brighter regions become dark.

Example:

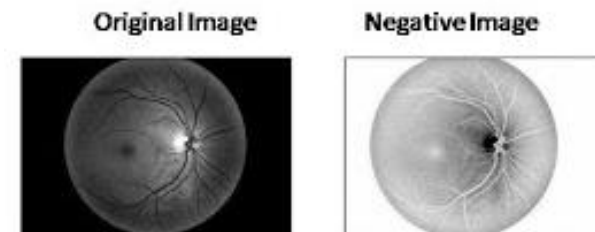
For an 8-bit image, $L = 256$.

Input intensity $r = 100$:

Output intensity $s = 256 - 1 - 100 = 155$.

Original Image			Negative Image		
0	255	127	255	0	128
10	250	50	245	5	205
200	100	50	55	155	205

8-bit & (3 X 3) Image





(a) Original



(b) Negative image obtained using the negative transformation

2.2 Logarithmic Transformations

These transformations **enhance the lower intensity values** in an image while **compressing the higher intensity values**.

a) Log Transformation:

$$c = \frac{255}{\log(1 + \max(r))}$$

Increases the visibility of dark regions by **mapping low intensity values** into a **broader range**.

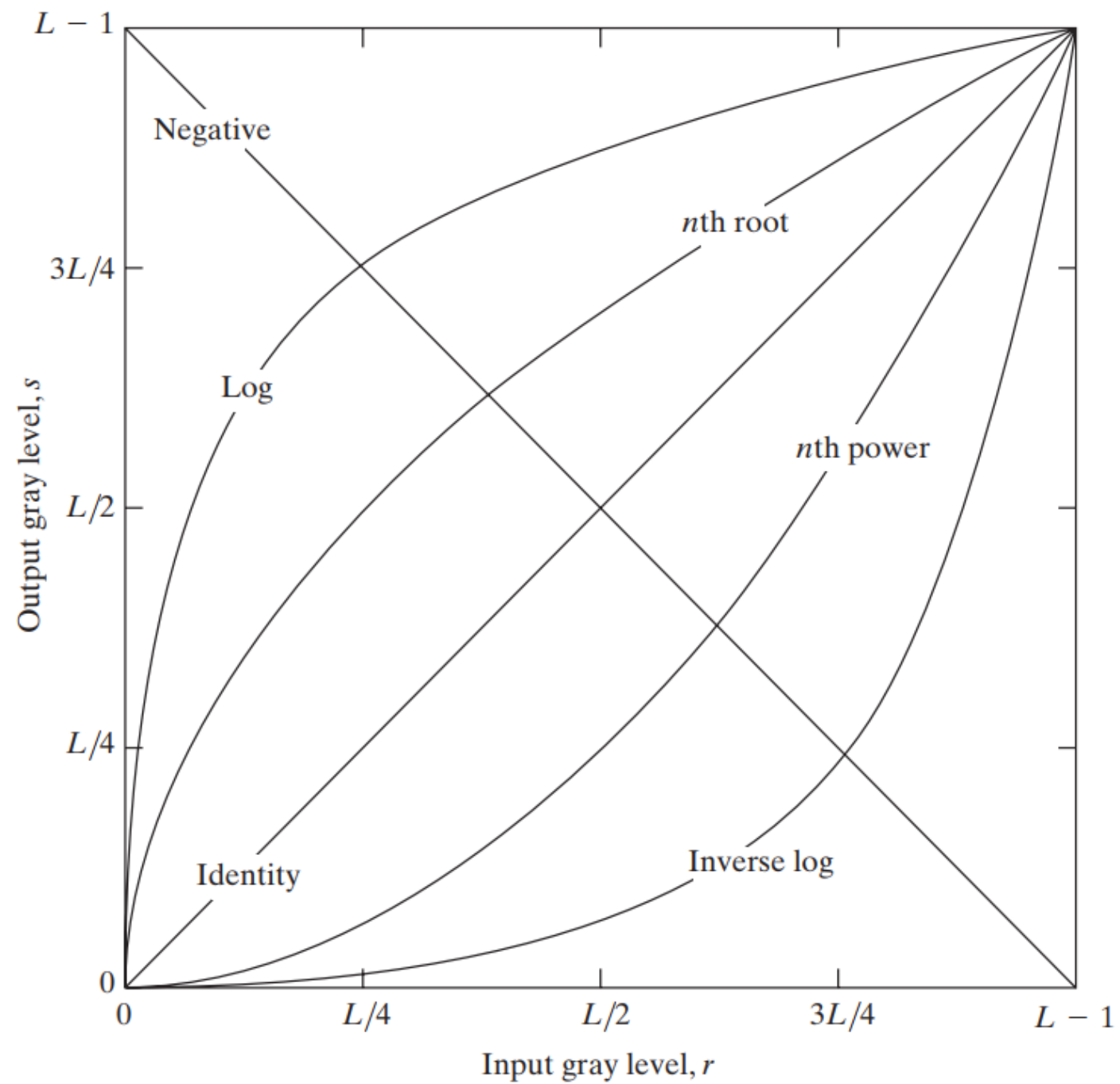
Equation: $s = c \cdot \log(1+r)$

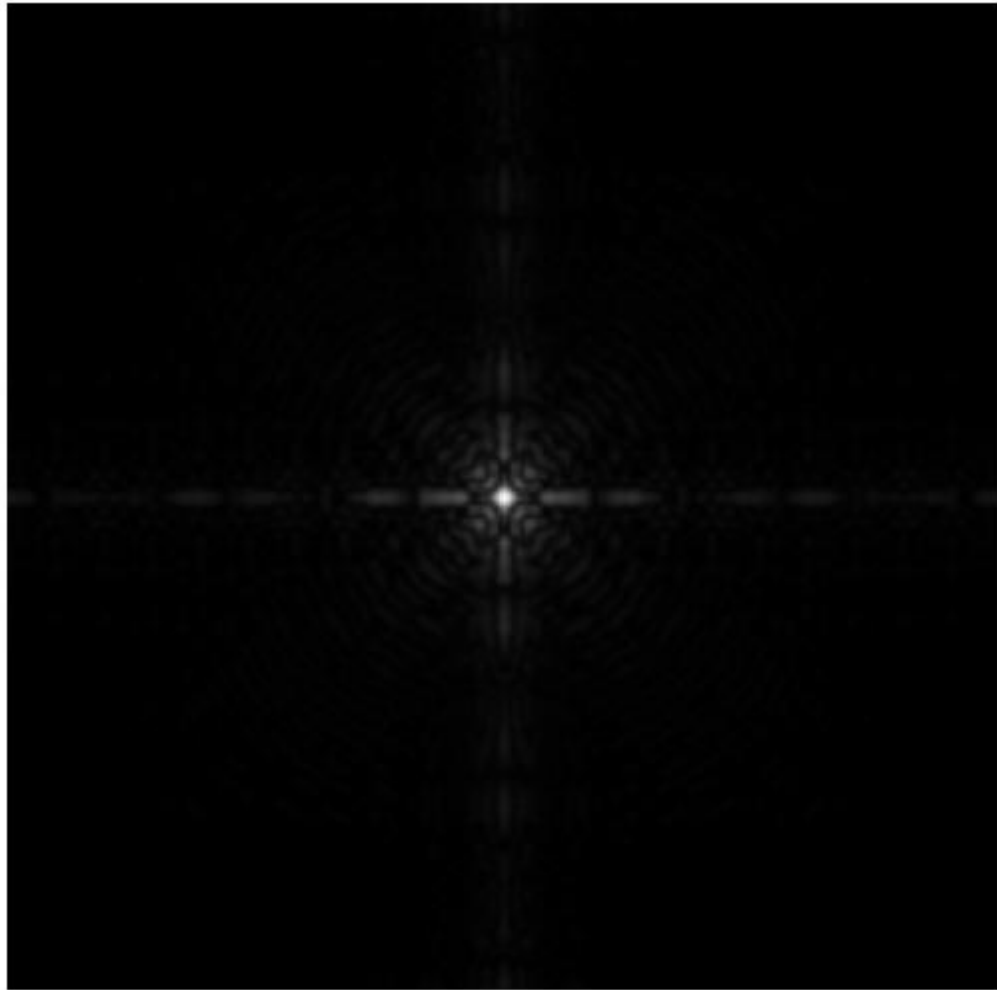
Where, c: A scaling constant.

If $r = 10$ and $c = 30$:

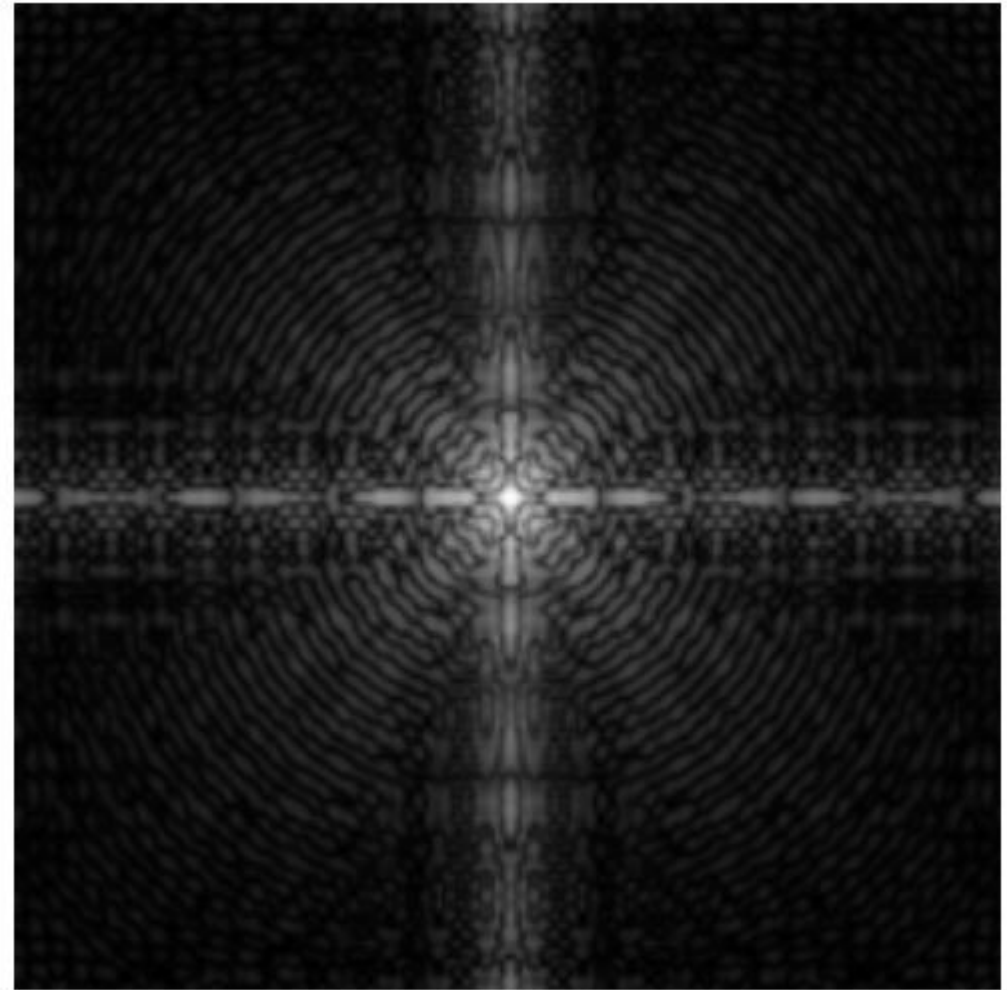
$$s = 30 \cdot \log(1 + 10) \approx 69.08.$$

Effect: Useful for displaying details in images with high dynamic ranges, like **astronomical or medical images**.





(a) Fourier spectrum.



(b) Result of applying the log transformation with $c=1$

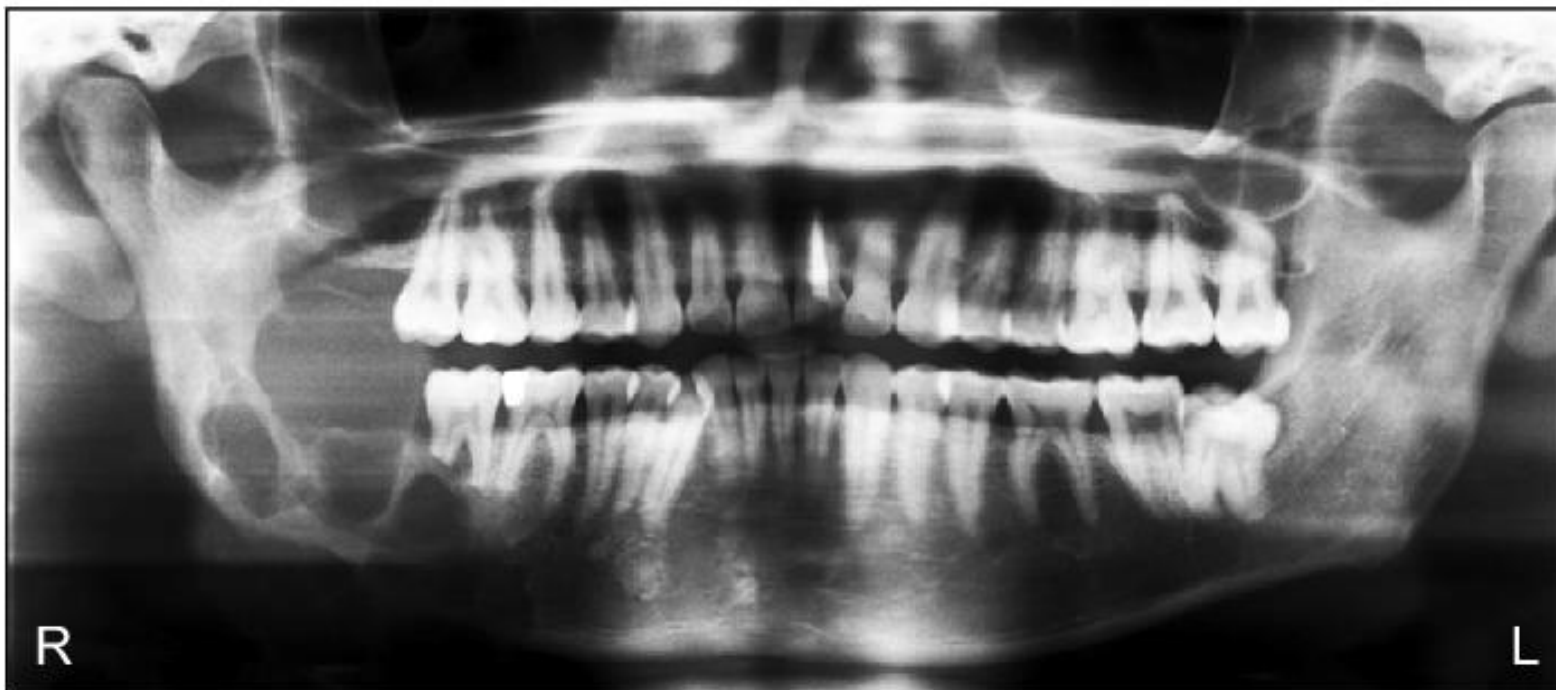
b) Inverse Log Transformation

Maps high intensity values to a wider range, reversing the effect of log transformation.

Equation: $s = e^{(r/c)} - 1$

$$c' = \frac{255}{\exp(\max(r)) - 1}$$

Effect: Useful for emphasizing brighter areas.



2.3 Power-Law (Gamma) Transformations

Also known as **gamma correction**, this transformation is used for varying the brightness and contrast of an image.

a) nth Power Transformation:

Equation: $s = c \cdot r^\gamma$

If $r = 0.5$, $c = 1$, and $\gamma = 2$:

$$s = 1 \cdot (0.5)^2 = 0.25.$$

c : Scaling constant.

γ : The gamma value that controls the transformation.

Effect: If $\gamma < 1$: Brightens the image.

If $\gamma > 1$: Darkens the image.

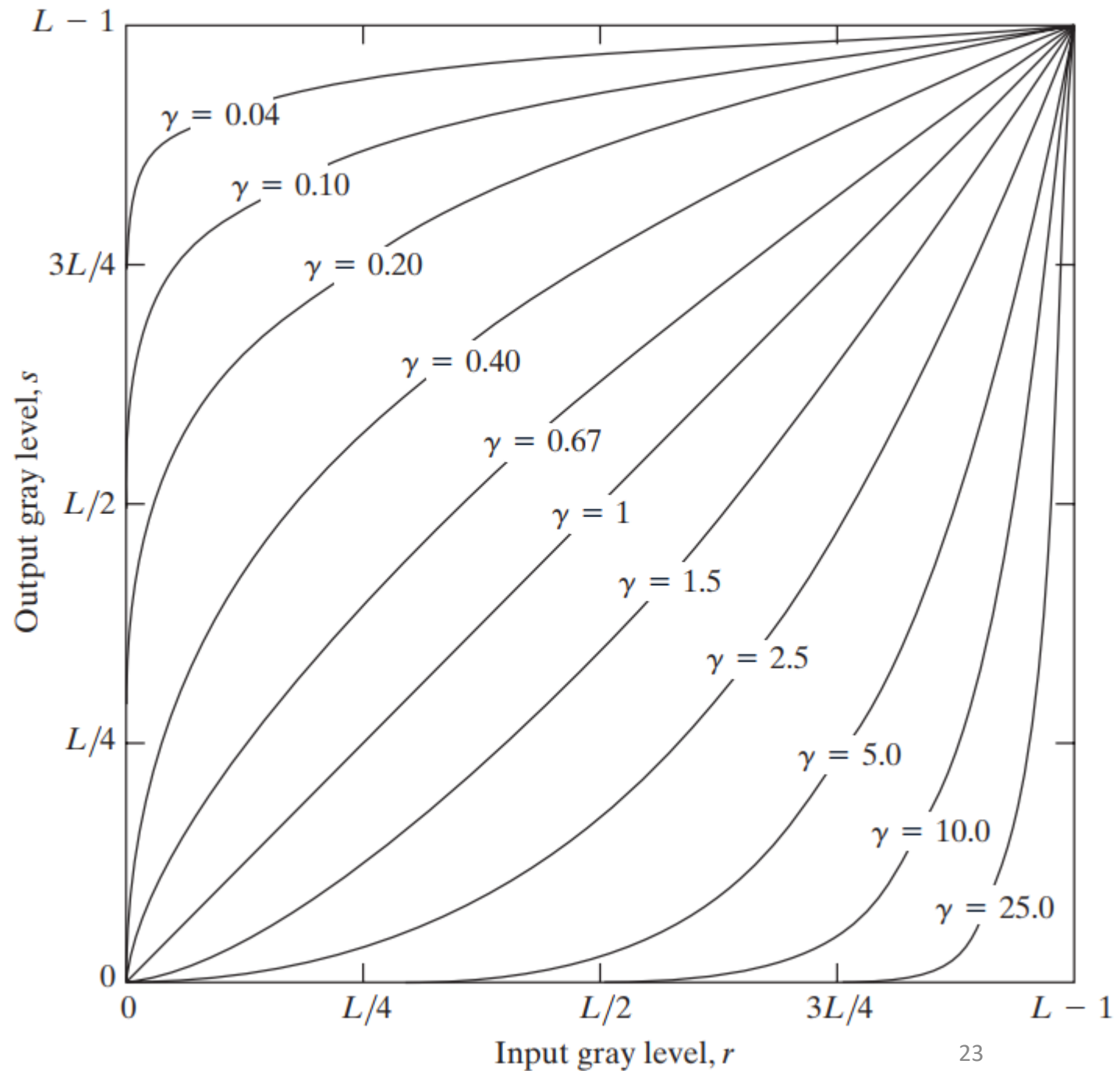
b) nth Root Transformation

Equation:

$$s = c \cdot r^{1/\gamma}$$

Effect: The reverse of nth power, emphasizing brighter regions.

$$c = \frac{255}{\max(r)^{\frac{1}{n}}}$$



What is Gamma Correction?

In CRTs, the intensity of light emitted is not directly proportional to the input voltage. Instead, it follows a power-law relationship.

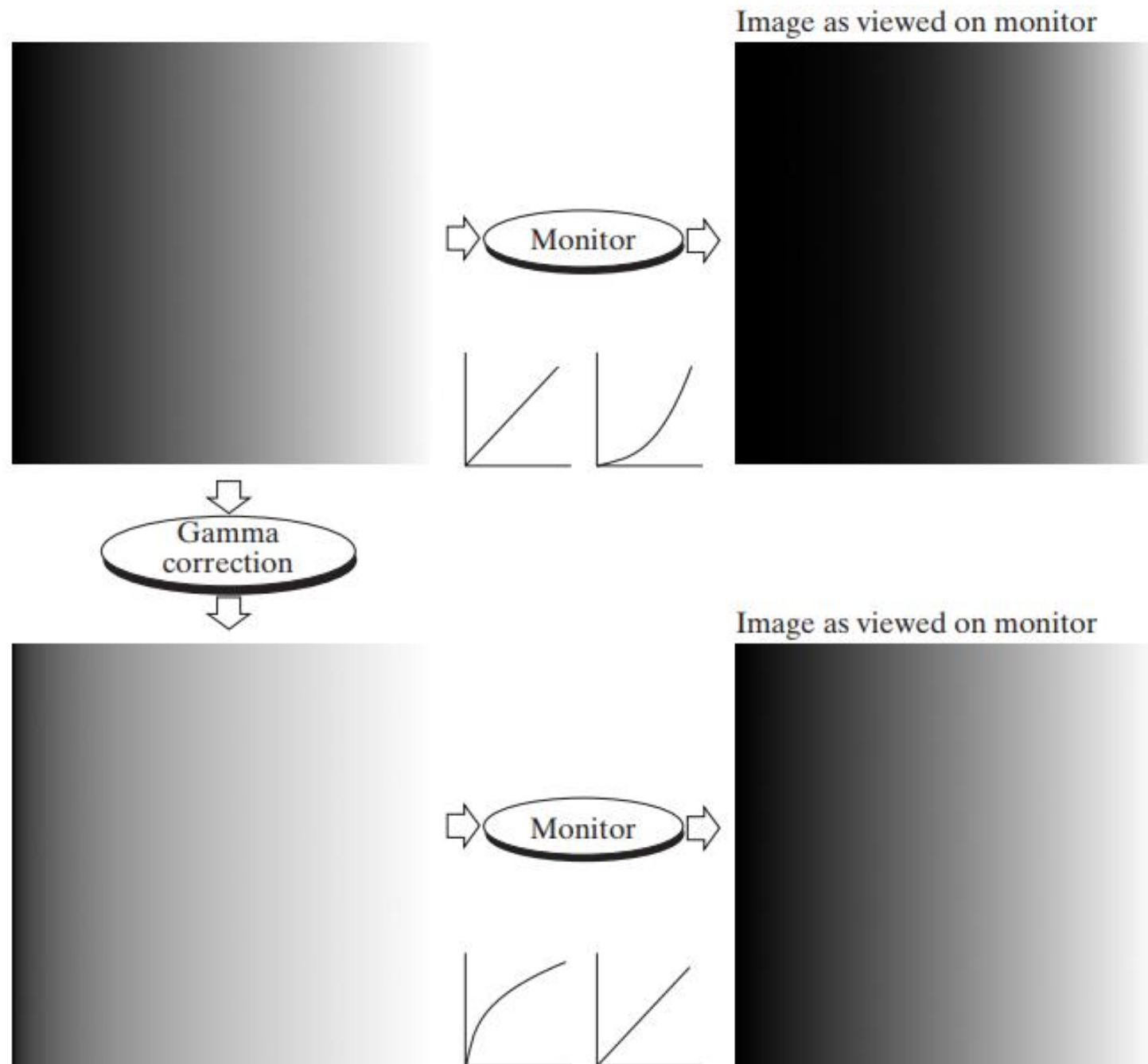
$$I = V^\gamma$$

I : Light intensity emitted by the screen.

V : Input voltage.

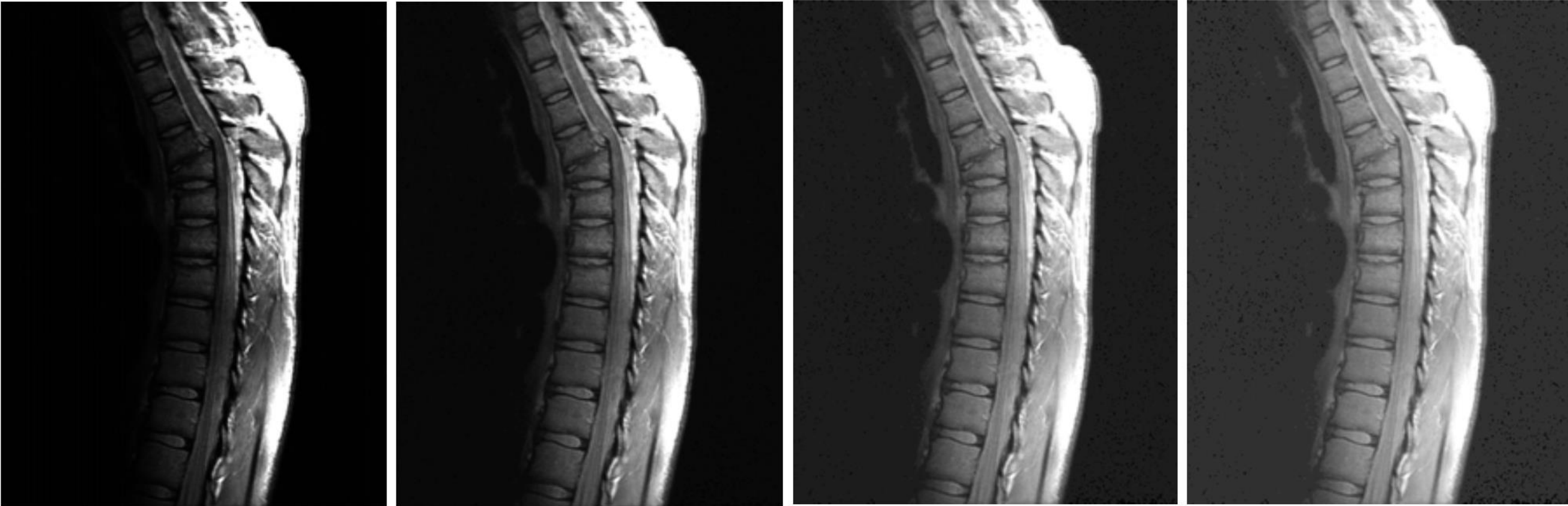
γ : The gamma value, typically around 2.2 for CRT monitors.

$$I_{\text{corrected}} = I_{\text{input}}^{1/\gamma}$$



Applications of Gamma Correction

(a) Magnetic resonance (MR) image of a fractured human spine.



(a)

(b)

(c)

(d)

(b)–(d) Results of applying the transformation in Eq. (3.2-3) with $c=1$ and $\gamma = 0.6, 0.4, \text{ and } 0.3$

Aerial image



Corrected image(Gamma=3)



Corrected image(Gamma=4)



Corrected image(Gamma=5)



3. Piecewise-Linear Transformation Functions

3.1 Contrast stretching

Low-contrast images can result from

- ❖ poor illumination,
- ❖ lack of dynamic range in the imaging sensor, or even
- ❖ wrong setting of a lens aperture during image acquisition.

The idea behind contrast stretching is to increase the dynamic range of the gray levels in the image being processed.

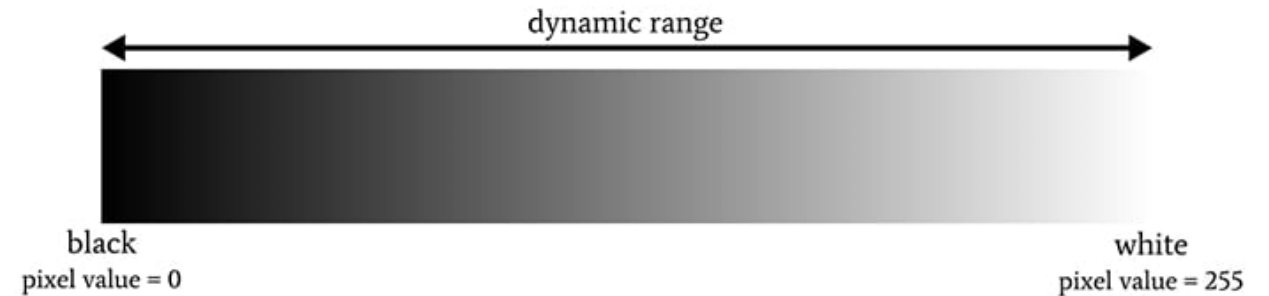
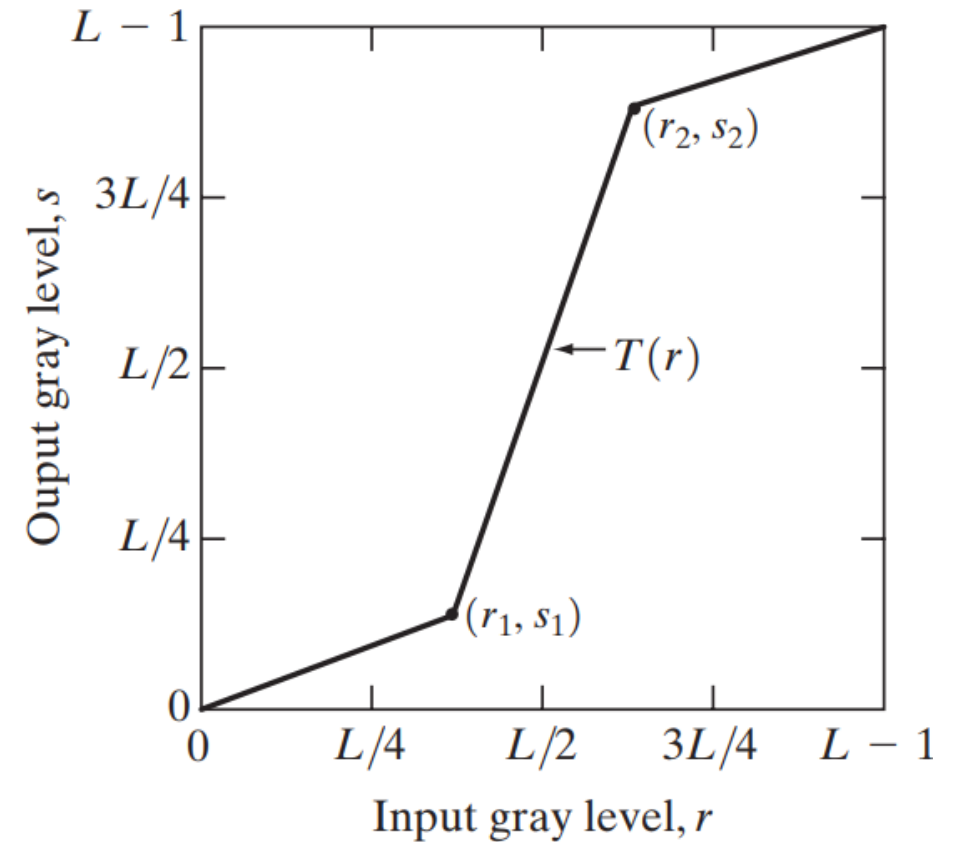


Figure shows a typical transformation used for contrast stretching.

The locations of points (r_1, s_1) and (r_2, s_2) control the shape of the transformation function.

- ✓ If $r_1=s_1$ and $r_2=s_2$, the transformation is a linear function that produces no changes in gray levels.
- ✓ If $r_1=r_2$, $s_1=0$ and $s_2=L-1$, the transformation becomes a thresholding function that creates a binary image
- ✓ Intermediate values of (r_1, s_1) and (r_2, s_2) produce various degrees of spread in the gray levels of the output image, thus affecting its contrast.



Form of transformation function

$$s = s_1 + \frac{(r - r_1)}{(r_2 - r_1)} \cdot (s_2 - s_1)$$

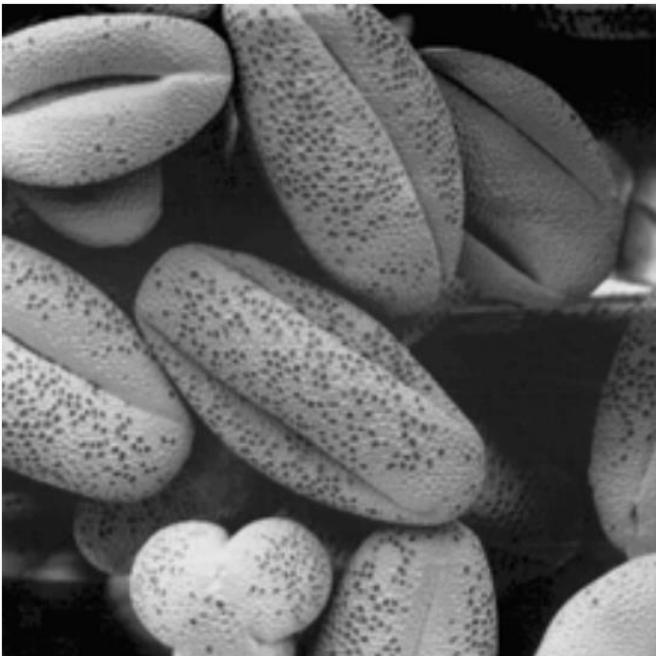
r : Input pixel intensity.

r_1, r_2 : Minimum and maximum intensities in the input range to be stretched.

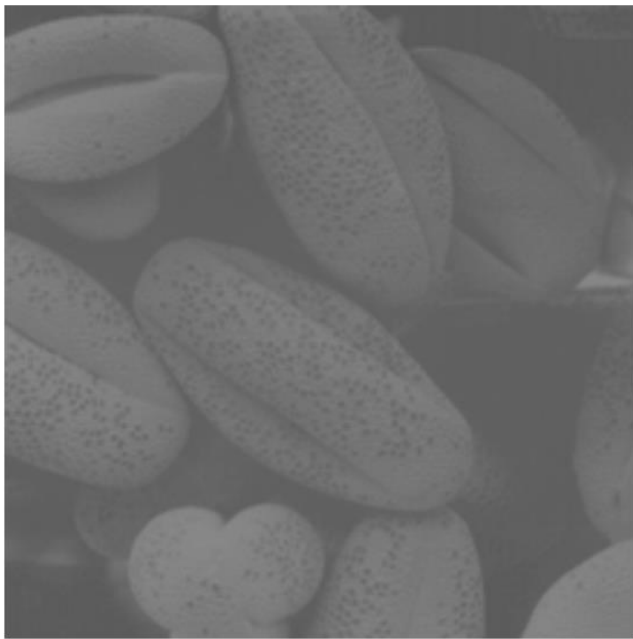
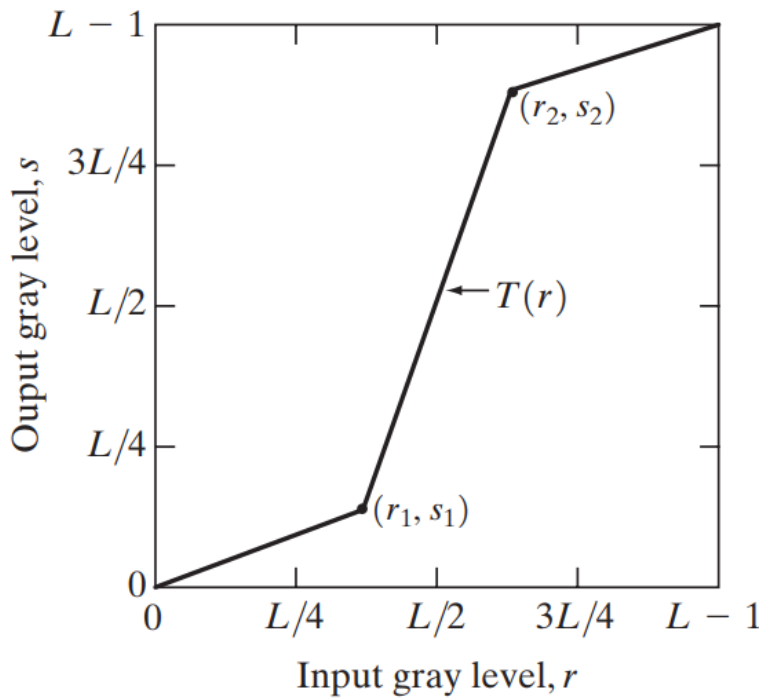
s : Output pixel intensity.

s_1, s_2 : Minimum and maximum intensities in the desired output range.

Result of contrast stretching



Form of transformation function



A low-contrast image.

Result of thresholding



$$(r_1, s_1) = (r_{\min}, 0)$$

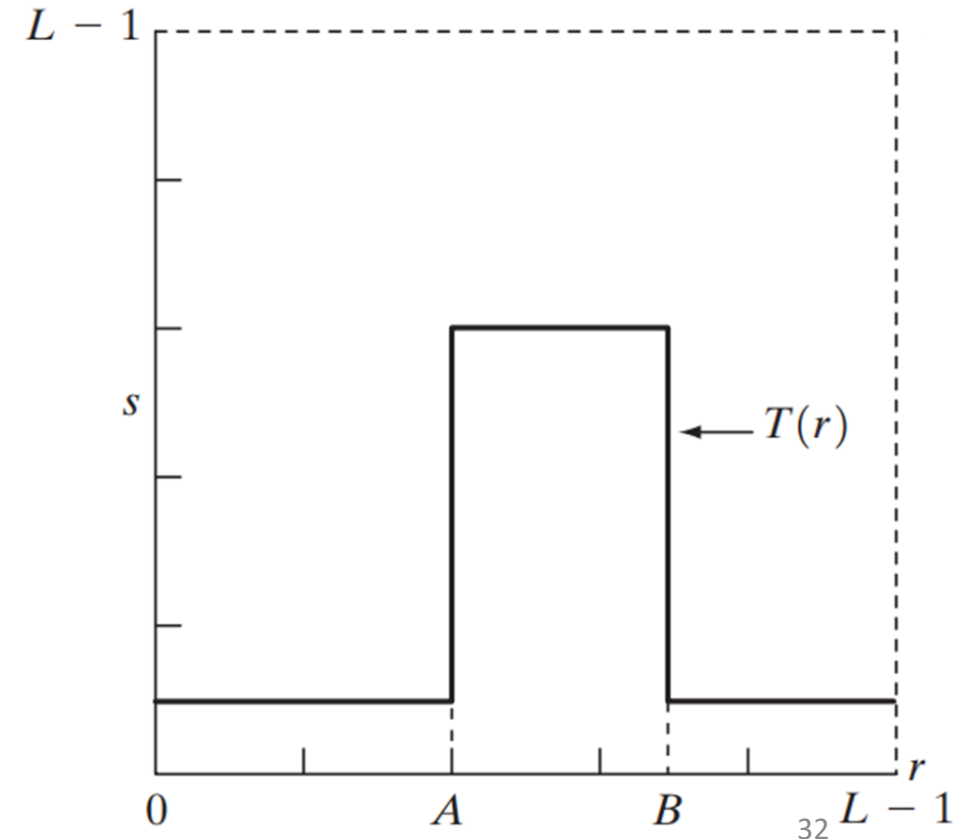
$$(r_2, s_2) = (r_{\max}, L - 1)$$

3.2 Gray-level slicing

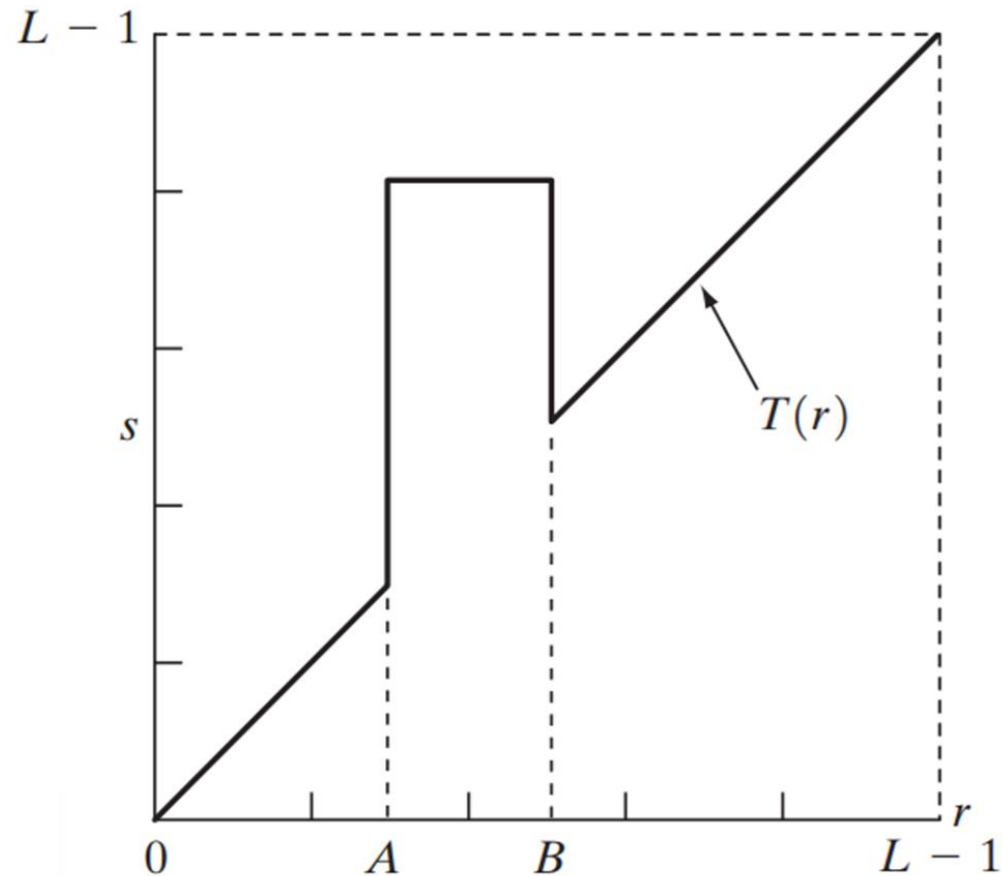
Highlighting a specific range of gray levels in an image often is desired.

- ❑ One approach is to display a high value for all gray levels in the range of interest and a low value for all other gray levels.

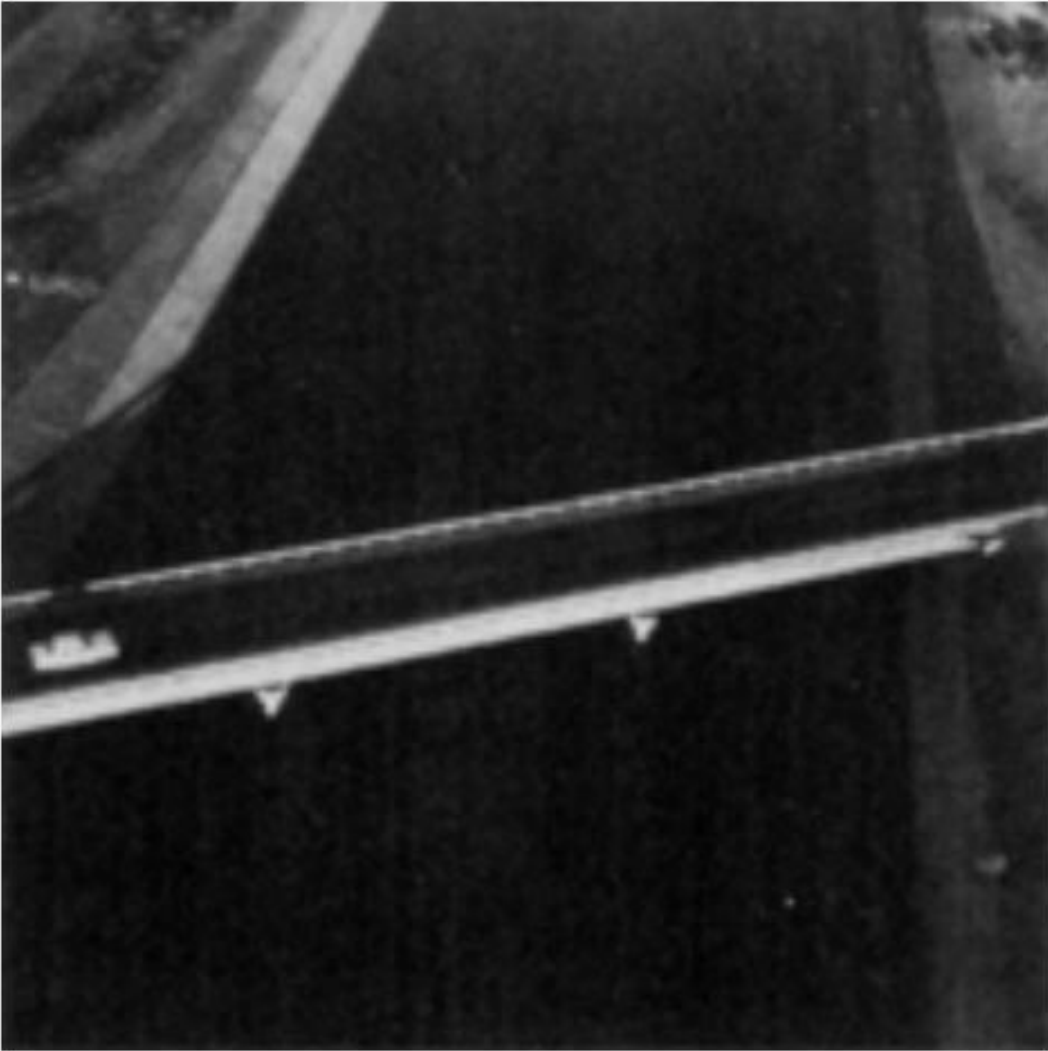
This transformation highlights range $[A, B]$ of gray levels and reduces all others to a constant level.



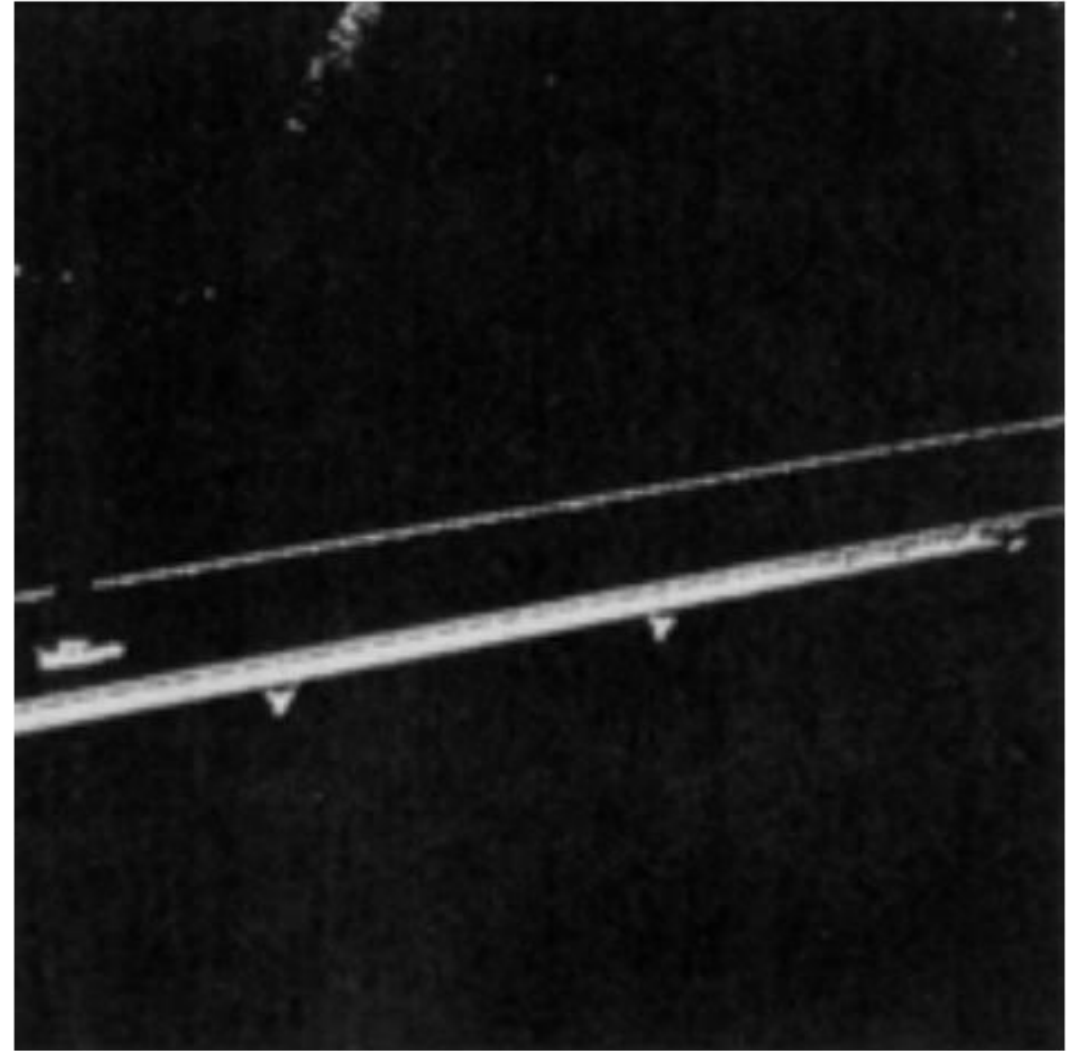
- ❑ The second approach: brightens the desired range of gray levels but preserves the background and gray-level tonalities in the image.



This transformation highlights range [A, B] but preserves all other levels.

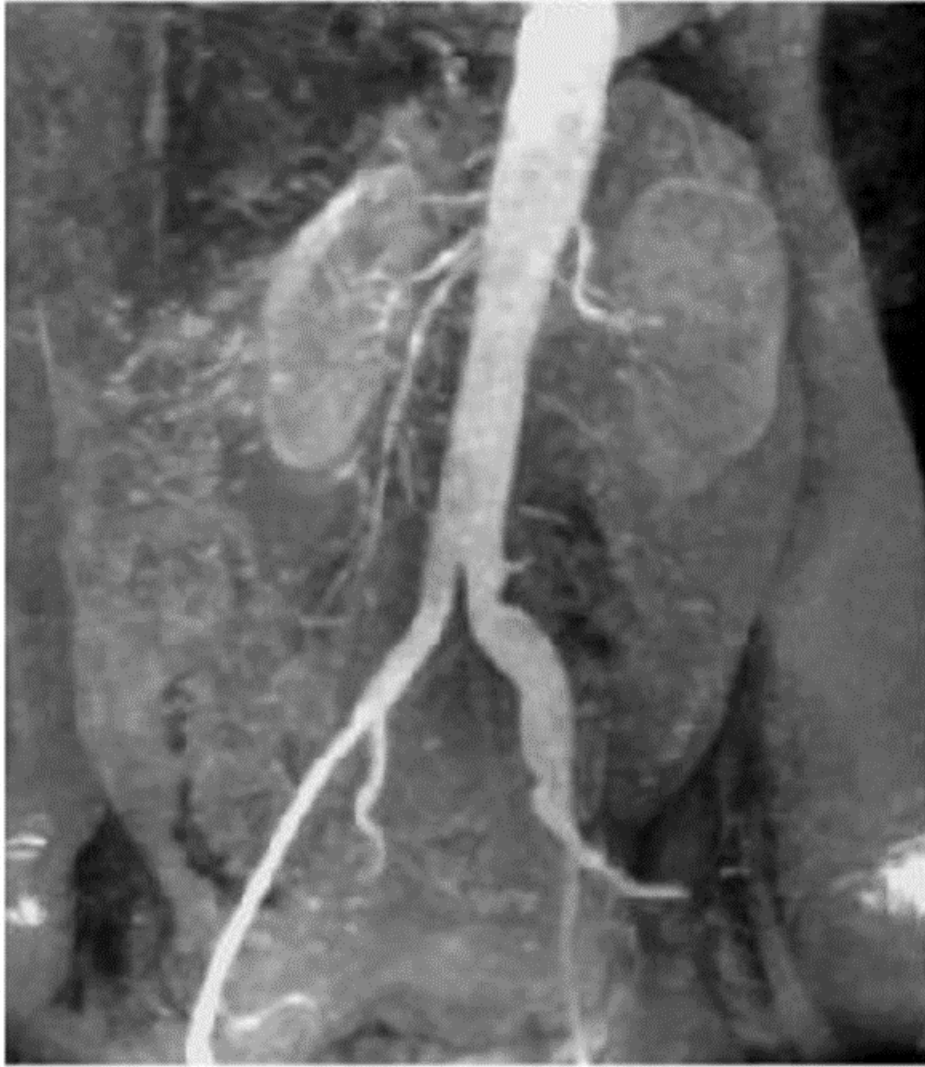


Result of using the transformation
in second approach



Result of using the transformation
in first approach

Original image



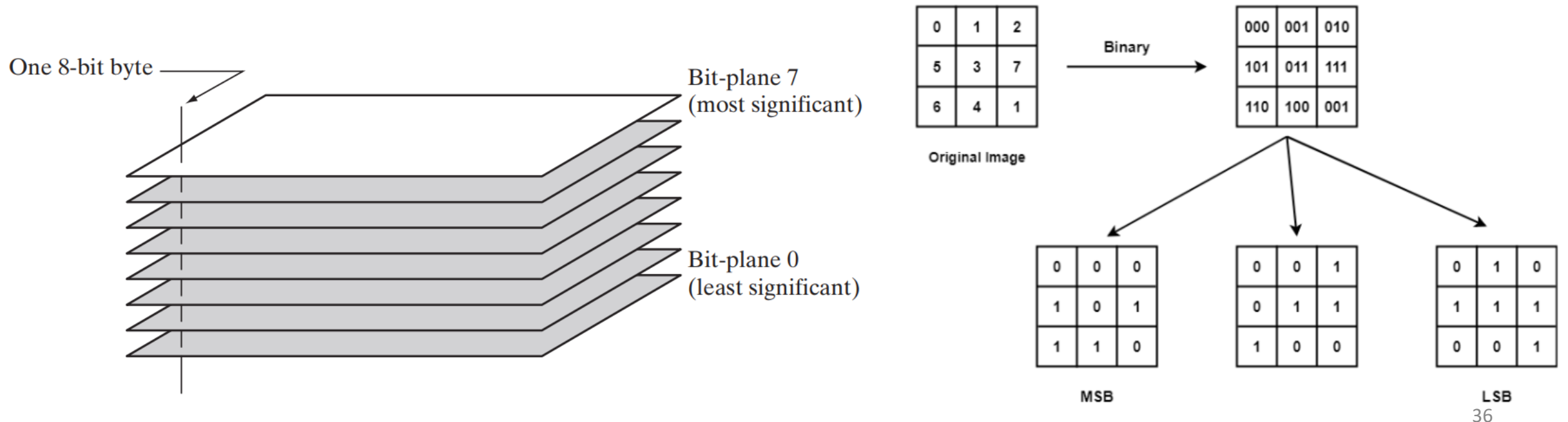
Graylevel slicing without background



3.3 Bit-plane slicing

Bit-plane slicing is a technique used in image processing to **analyze the contribution of individual bits** in the representation of an image.

It involves separating an image into binary layers (bit-planes) based on the binary representation of the pixel values.



3.3.1 Pixel Representation:

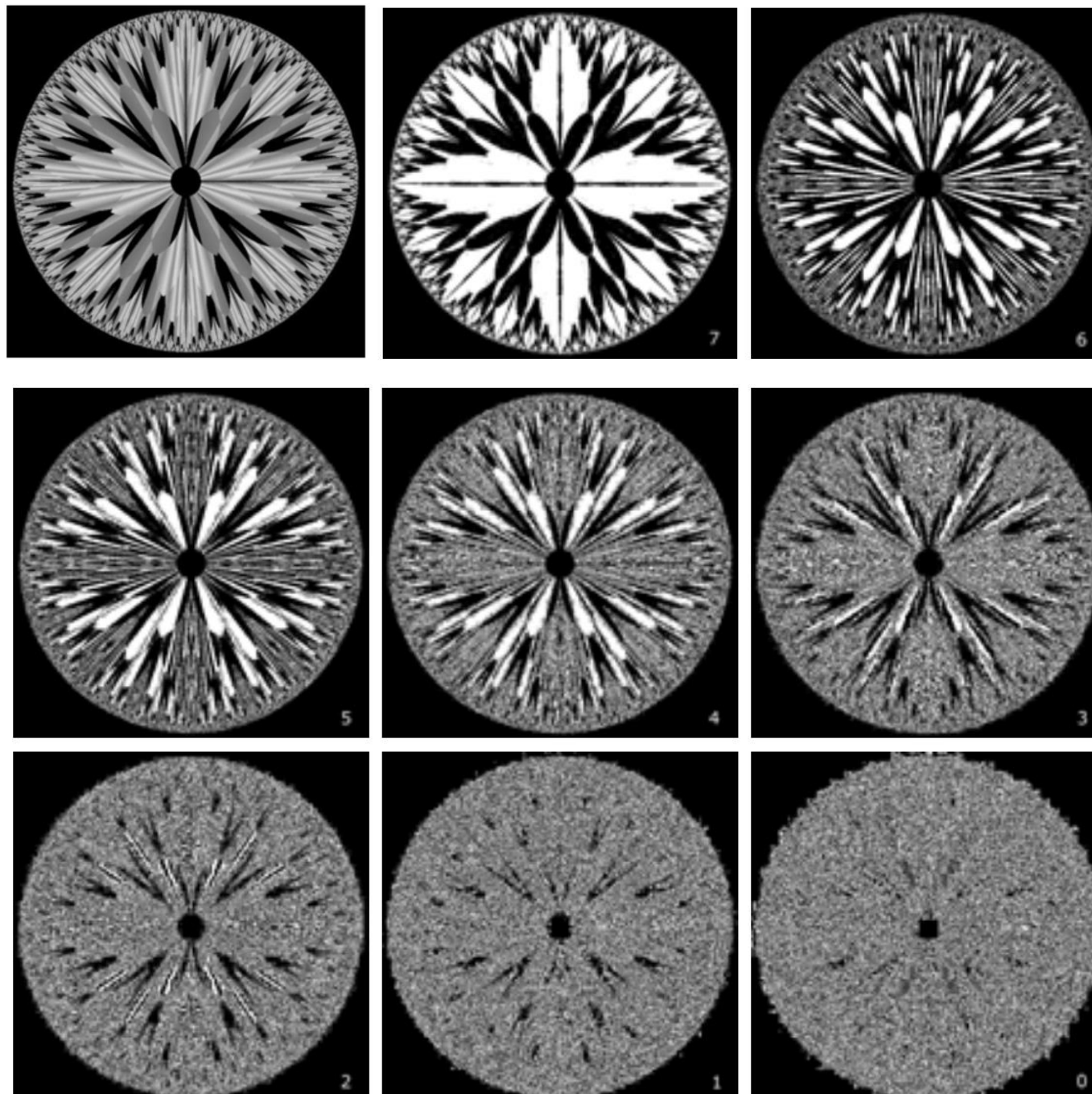
Each pixel value in an 8-bit grayscale image is represented by a binary number with 8 bits (e.g., 10010110 for a pixel with intensity 150).

Each of these 8 bits contributes differently to the overall intensity.

Bit-Planes:

Plane 0 (Least Significant Bit - LSB): Contains the lowest-order bits, representing the smallest contribution to the pixel's intensity.

Plane 7 (Most Significant Bit - MSB): Contains the highest-order bits, representing the largest contribution to the pixel's intensity.



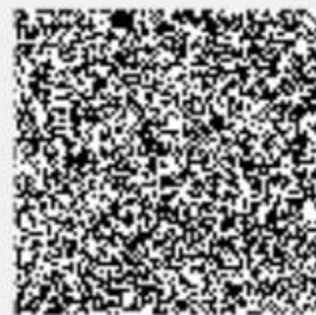
R Bit Slicing 1



R Bit Slicing 2



R Bit Slicing 3



R Bit Slicing 4



R Bit Slicing 5



R Bit Slicing 6



R Bit Slicing 7



R Bit Slicing 8



R Channel Image

